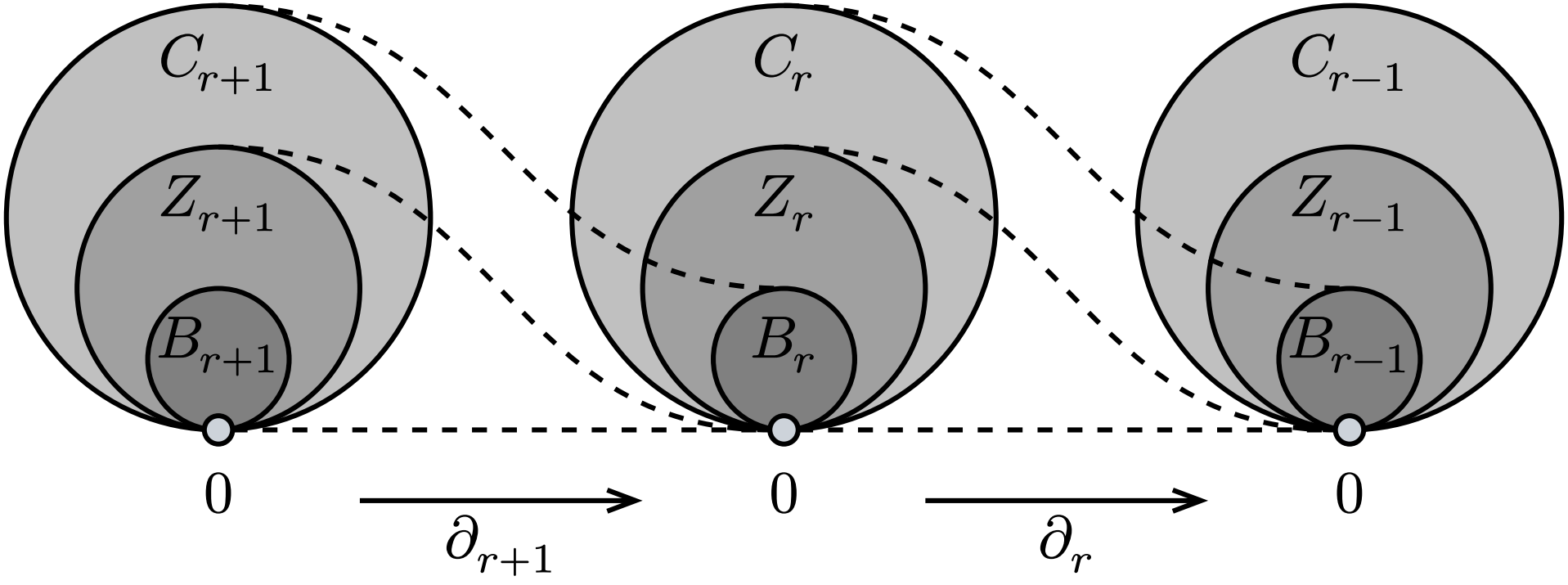


$C_r$: $r$-chains, $Z_r = \ker \partial_r$: $r$-cycles

$B_r = \operatorname{im} \partial_{r+1}$: $r$-boundaries, $\partial_r$: boundary map

homology group $H_r = Z_r / B_r$